Review plan · for the lecturer

Teaching AI security at university: unit check

AI security · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Adversarial examples
- Poisoning and backdoors
- Privacy attacks
- Prompt injection

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you keep red teaming safe?

Exit ticket

- One change to my AI security course this term, and how I will know it worked.

Try it in your next lecture
Bring your attack-the-defence assignment and red-team scope document to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **FGSM moves each feature by ε in the direction of**

◯ A random sign ◯ The sign of the loss gradient ◯ The negative of the input

2 **Bound the change.**

w = (5, -2, 1), ε = 0.1: the score can change by _____ .

Word bank: 0.5 · 0.8 · 0.1 · 0.4

3 **A backdoored model usually shows low clean accuracy.**

◯ True ◯ False

4 **The strongest evidence that a defence is robust:**

◯ High clean accuracy ◯ FGSM fails against it

◯ It survives adaptive attacks under a stated threat model

5 **Set the noise.**

Sensitivity 1, ε = 0.25: Laplace scale = _____ .

Word bank: 2 · 4 · 0.25

6 **Which increase membership inference risk? Choose all that apply.**

☐ Releasing full confidence scores ☐ A large train-test gap ☐ Training with DP-SGD

7 **Hidden instructions in a retrieved web page are**

◯ Direct prompt injection ◯ Indirect prompt injection ◯ Data poisoning

8 **Match each attack to its stage.**

Poisoning	• • • •	Inference input
Adversarial example	• • • •	Model outputs
Membership inference	• • • •	Training

9 **Publishing the same count twice, with fresh noise at ε = 0.5 each time, uses as much privacy budget as one release at ε = 1.**

◯ True ◯ False

10 **Best way to assess a student red-team exercise:**

◯ Number of jailbreaks ◯ Attacks on live systems

◯ Quality of findings and mitigations within scope

Answers · for the lecturer

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1. **The sign of the loss gradient**
   It increases the loss fastest per feature.

2. **w = (5, −2, 1), ε = 0.1: the score can change by 0.8.**
   ε × Σ|w| = 0.1 × (5 + 2 + 1) = 0.8. The signed sum 4 would give 0.4, and the largest weight alone 0.5: every feature moves, so every $|w_i|$ counts.

3. **False**
   Backdoors preserve clean accuracy.

4. **It survives adaptive attacks under a stated threat model**
   Attack the defence itself.

5. **Sensitivity 1, ε = 0.25: Laplace scale = 4.**
   Scale = Δ/ε = 1 / 0.25 = 4: the smaller ε, the more noise.

6. **A large train-test gap; Releasing full confidence scores**
   DP-SGD reduces the risk.

7. **Indirect prompt injection**
   The attack arrives through content.

8. **Poisoning — Training**
   Adversarial example — Inference input
   Membership inference — Model outputs
   Each stage needs its own controls.

9. **True**
   Repeating a query isn't free: each fresh release spends ε = 0.5, and basic composition adds them up to ε = 1.

10. **Quality of findings and mitigations within scope**
    Reward analysis, not damage.